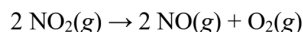


Kinetics

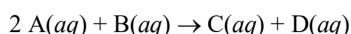
1. Nitrogen dioxide decomposes to nitric oxide and oxygen according to the following reaction:



If $\Delta[\text{NO}_2]/\Delta t$ is -0.0024 M s^{-1} , what is $\Delta[\text{O}_2]/\Delta t$?

- (A) -0.0048 M s^{-1} (B) -0.0012 M s^{-1}
(C) 0.0048 M s^{-1} (D) 0.0012 M s^{-1}

2. For the reaction below, initial rate data appear in the table. What is the rate law for the reaction?



Experiment	$[\text{A}]_0, \text{M}$	$[\text{B}]_0, \text{M}$	Initial rate, M s^{-1}
1	0.10	0.10	0.086
2	0.20	0.10	0.342
3	0.20	0.20	0.344

- (A) $\text{Rate} = k[\text{A}][\text{B}]$ (B) $\text{Rate} = k[\text{A}]^2$
(C) $\text{Rate} = k[\text{A}]^2[\text{B}]$ (D) $\text{Rate} = k[\text{A}][\text{B}]^2$

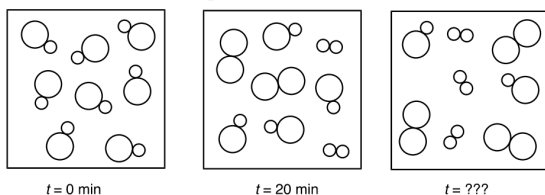
3. Two reactions have the same rate constant at 300.0 K. Reaction 1 has an activation energy of 48.2 kJ/mol, while reaction 2 has an activation energy of 20.0 kJ/mol. What is k_1/k_2 at 320.0 K?

- (A) 0.493 (B) 0.844
(C) 1.18 (D) 2.03

4. The reaction between A and B has the rate law $\text{Rate} = k[\text{A}][\text{B}]$. A reaction is conducted with $[\text{A}]_0 = 0.0245 \text{ M}$ and $[\text{B}]_0 = 3.24 \times 10^{-5} \text{ M}$, and the natural logarithm of the concentration of [B] is plotted as a function of time in seconds. The plot is linear, with a slope of -3.96×10^{-4} . What is the value of k ?

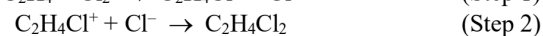
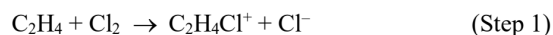
- (A) $3.96 \times 10^{-4} \text{ M}^{-1} \text{ s}^{-1}$ (B) $1.62 \times 10^{-2} \text{ M}^{-1} \text{ s}^{-1}$
(C) $12.2 \text{ M}^{-1} \text{ s}^{-1}$ (D) $499 \text{ M}^{-1} \text{ s}^{-1}$

5. In the depiction of an irreversible chemical reaction shown below, the large circles represent atoms of element A and the small circles represent atoms of element B. At $t = 0$, only compound AB is present. The rate law of the reaction is $\text{Rate} = k[\text{AB}]^2$. What is the value of t in the third panel?



- (A) 30 min (B) 40 min
(C) 50 min (D) 60 min

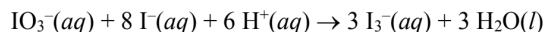
6. The chlorination of ethene is proposed to occur by the following mechanism:



Which step could possibly be rate-determining, given a rate law of $\text{Rate} = k[\text{C}_2\text{H}_4][\text{Cl}_2]$?

- (A) Step 1 only
(B) Step 2 only
(C) Either Step 1 or Step 2
(D) Neither Step 1 nor Step 2

7. When iodate ion reacts with iodide ion in acidic solution, which statement about how concentrations are changing is correct?



- (A) $8 \frac{\Delta[\text{I}^-]}{\Delta t} = 3 \frac{\Delta[\text{I}_3^-]}{\Delta t}$ (B) $8 \frac{\Delta[\text{I}^-]}{\Delta t} = -3 \frac{\Delta[\text{I}_3^-]}{\Delta t}$
(C) $3 \frac{\Delta[\text{I}^-]}{\Delta t} = 8 \frac{\Delta[\text{I}_3^-]}{\Delta t}$ (D) $3 \frac{\Delta[\text{I}^-]}{\Delta t} = -8 \frac{\Delta[\text{I}_3^-]}{\Delta t}$

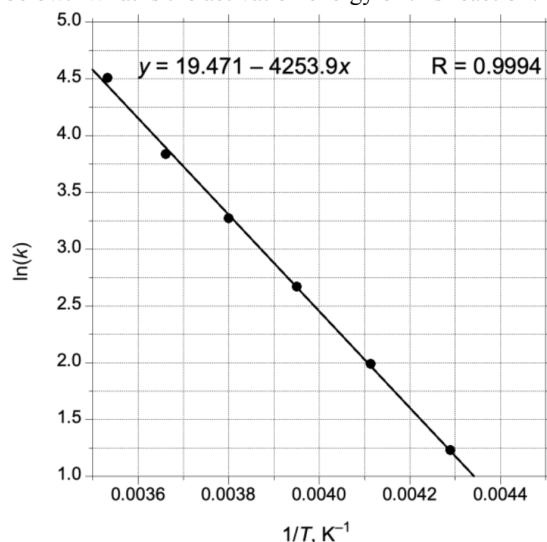
8. For the irreversible reaction between $\text{CO}(\text{g})$ and $\text{Cl}_2(\text{g})$ to form $\text{COCl}_2(\text{g})$, the rate law is $\text{Rate} = k[\text{CO}][\text{Cl}_2]$. In an experiment where equimolar quantities of $\text{CO}(\text{g})$ and $\text{Cl}_2(\text{g})$ are reacted, which of the following would change linearly with reaction time?

- (A) $\ln([\text{CO}])$ (B) $\ln([\text{COCl}_2])$
(C) $1/[\text{CO}]$ (D) $1/[\text{COCl}_2]$

9. A sample containing radium-226 was measured in 1925 to have an activity of 3.74×10^6 disintegrations per second. The same sample, re-measured in 2025, has an activity of 3.58×10^6 disintegrations per second. What is the half-life of ^{226}Ra ?

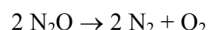
- (A) 620 years (B) 1200 years
(C) 1600 years (D) 2300 years

10. The rate constant for the addition of pyridine to an iridium compound is measured at a variety of temperatures, and the variation of $\ln(k)$ with $1/T$ is shown below. What is the activation energy of this reaction?

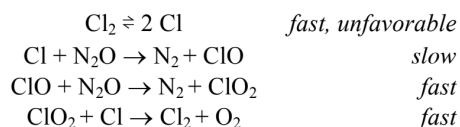


- (A) 1.82 kJ mol^{-1} (B) 4.25 kJ mol^{-1}
 (C) 35.4 kJ mol^{-1} (D) 82.8 kJ mol^{-1}
11. Cyclopropane is proposed to isomerize to propene through an intermediate diradical as shown below. The overall reaction is observed to follow the rate law $\text{Rate} = k[\text{cyclo-C}_3\text{H}_6]$. If k_2 is much smaller than k_{-1} , what is the value of k in terms of k_1 , k_{-1} , and k_2 ?
- $$\text{cyclo-C}_3\text{H}_6 \xrightleftharpoons[k_{-1}]{k_1} \cdot\text{CH}_2\text{CH}_2\text{CH}_2\cdot$$
- $$\cdot\text{CH}_2\text{CH}_2\text{CH}_2\cdot \xrightarrow{k_2} \text{CH}_3\text{CH}=\text{CH}_2$$
- (A) k_1 (B) k_1/k_{-1}
 (C) k_1k_2 (D) k_1k_2/k_{-1}
12. The reaction of $\text{MnO}_4^-(aq)$ and $\text{H}_2\text{C}_2\text{O}_4(aq)$ in acidic solution is autocatalytic because one of the products, $\text{Mn}^{2+}(aq)$, is a reducing agent. Starting from a mixture of reactants, which of the following graphs best shows how the concentration of $\text{MnO}_4^-(aq)$ changes over time?
- (A) (B) (C) (D)
13. Ethanol ($\text{C}_2\text{H}_5\text{OH}$) is oxidized to acetaldehyde (CH_3CHO) by dichromate ions in acidic solution according to the following equation:
- $$3 \text{C}_2\text{H}_5\text{OH}(aq) + \text{Cr}_2\text{O}_7^{2-}(aq) + 8 \text{H}^+(aq) \rightarrow 3 \text{CH}_3\text{CHO}(aq) + 2 \text{Cr}^{3+}(aq) + 7 \text{H}_2\text{O}(l)$$
- If the concentration of $\text{Cr}^{3+}(aq)$ is increasing at a rate of 0.18 M s^{-1} , at what rate is the concentration of $\text{CH}_3\text{CHO}(aq)$ changing?
- (A) Increasing at 0.12 M s^{-1}
 (B) Increasing at 0.27 M s^{-1}
 (C) Decreasing at 0.12 M s^{-1}
 (D) Decreasing at 0.27 M s^{-1}
14. A reaction in the gas phase doubles in rate on increasing the temperature from 10°C to 20°C . Which is the best explanation for the increase in reaction rate?
- (A) The pressure doubles.
 (B) The number of particles above a minimum activation energy doubles.
 (C) The average kinetic energy of the molecules doubles.
 (D) The number of collisions per unit time doubles.
15. An irreversible reaction occurs as follows:
- $$2 \text{A}(g) + \text{B}(g) \rightarrow \text{C}(g) + \text{D}(g)$$
- What is the rate law for this reaction?
- (A) $\text{Rate} = k[\text{A}][\text{B}]$
 (B) $\text{Rate} = k[\text{A}]^2[\text{B}]$
 (C) $\text{Rate} = k[2\text{A}][\text{B}]$
 (D) It cannot be determined from the information given.
16. The half-life of ^{90}Sr is 28 years. How long is required for the level of radioactivity to drop to 1.5% of its initial value?
- (A) 0.42 years (B) 40 years
 (C) 55 years (D) 170 years
17. An uncatalyzed reaction has an activation energy $E_a = 50 \text{ kJ mol}^{-1}$ and an enthalpy change $\Delta H^\circ = -10 \text{ kJ mol}^{-1}$. In the presence of a catalyst, what might these values become?
- (A) $E_a = 30 \text{ kJ mol}^{-1}$, $\Delta H^\circ = -10 \text{ kJ mol}^{-1}$
 (B) $E_a = 50 \text{ kJ mol}^{-1}$, $\Delta H^\circ = -15 \text{ kJ mol}^{-1}$
 (C) $E_a = 60 \text{ kJ mol}^{-1}$, $\Delta H^\circ = -20 \text{ kJ mol}^{-1}$
 (D) $E_a = -10 \text{ kJ mol}^{-1}$, $\Delta H^\circ = 50 \text{ kJ mol}^{-1}$

18. Chlorine gas catalyzes the gas-phase decomposition of nitrous oxide according to the equation



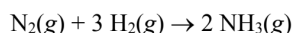
The following mechanism is proposed for the catalyzed reaction:



What rate law is predicted by this mechanism?

- (A) $\text{Rate} = k[\text{N}_2\text{O}]$ (B) $\text{Rate} = k[\text{N}_2\text{O}]^2$
(C) $\text{Rate} = k[\text{N}_2\text{O}][\text{Cl}_2]$ (D) $\text{Rate} = k[\text{N}_2\text{O}][\text{Cl}_2]^{1/2}$

19. Hydrogen and nitrogen react according to the following equation:



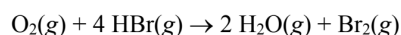
If the concentration of $\text{H}_2(\text{g})$ is changing at a rate of $-12 \text{ mol L}^{-1} \text{ s}^{-1}$, at what rate is the concentration of $\text{NH}_3(\text{g})$ changing?

- (A) $-6.0 \text{ mol L}^{-1} \text{ s}^{-1}$ (B) $-4.0 \text{ mol L}^{-1} \text{ s}^{-1}$
(C) $+8.0 \text{ mol L}^{-1} \text{ s}^{-1}$ (D) $+12 \text{ mol L}^{-1} \text{ s}^{-1}$

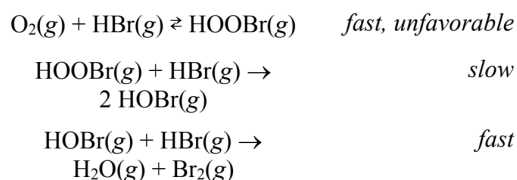
20. An irreversible reaction $\text{R} \rightarrow \text{P}$ has the rate law $\text{Rate} = k[\text{R}]^2$. Which graph will be a straight line?

- (A) $\ln([\text{R}])$ as a function of t
(B) $[\text{R}]$ as a function of $\ln(t)$
(C) $1/[\text{R}]$ as a function of t
(D) $[\text{R}]$ as a function of $1/t$

21. For the reaction



the following mechanism is proposed:



What rate law is predicted by this mechanism?

- (A) $\text{Rate} = k[\text{HBr}]$ (B) $\text{Rate} = k[\text{O}_2][\text{HBr}]$
(C) $\text{Rate} = k[\text{O}_2][\text{HBr}]^2$ (D) $\text{Rate} = k[\text{O}_2][\text{HBr}]^4$

22. The radioisotope $^{99\text{m}}\text{Tc}$ undergoes gamma decay with a rate constant $k = 0.116 \text{ h}^{-1}$. How much time is required for its activity to decrease to 10.0% of its original activity?

- (A) 19.8 h (B) 7.76 h (C) 5.98 h (D) 1.16 h

23. A cotton ball burns more brightly in a jar of 80% O_2 than in the air, which is 20% O_2 . Which is the best explanation for this observation?

- (A) The cotton collides more frequently with the O_2 inside the jar.
(B) The excited state of O_2 is formed more efficiently inside the jar.
(C) The temperature is higher inside the jar.
(D) The jar acts as a catalyst for the reaction.

24. A reaction has an activation energy of 75.0 kJ mol^{-1} . At what temperature will the rate constant be twice as large as it is at 298 K?

- (A) 305 K (B) 307 K (C) 311 K (D) 319 K

25. *Tert*-butyl chloride reacts with hydroxide ion in a process that is first order in both *tert*-butyl chloride and hydroxide. If both reactants are doubled in concentration, how does the reaction rate change?

- (A) The rate does not change.
(B) The rate doubles.
(C) The rate quadruples.
(D) The effect on the rate cannot be determined from the information given.

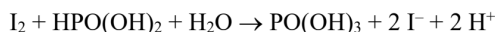
26. Under certain conditions, the rate of decomposition of hydrogen peroxide is first order in H_2O_2 . It is observed that the concentration of H_2O_2 decreases by 1.00% in 300.0 s. What is the rate constant for the reaction?

- (A) $3.35 \times 10^{-5} \text{ s}^{-1}$ (B) $2.31 \times 10^{-3} \text{ s}^{-1}$
(C) $3.33 \times 10^{-3} \text{ s}^{-1}$ (D) 300 s^{-1}

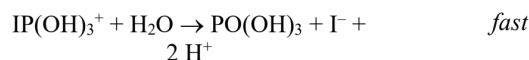
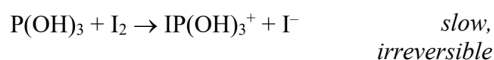
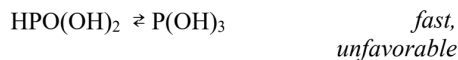
27. The pain reliever naproxen has a half-life of 12 h in the human body. A patient takes a 400 mg tablet of naproxen at 8 AM and a second 400 mg tablet at 8 PM (12 h after the first dose). How much naproxen remains in the patient's body at 8 AM the following day (12 h after the second dose)?

- (A) 50 mg (B) 100 mg
(C) 200 mg (D) 300 mg

28. For the aqueous reaction



the following mechanism is proposed:



Which observation is consistent with this mechanism?

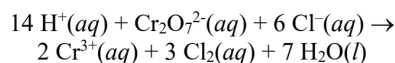
- (A) The reaction is first order in both $\text{HPO}(\text{OH})_2$ and I_2 .
 (B) The reaction rate is independent of the I_2 concentration.
 (C) The reaction goes faster as the concentration of H^+ is increased.
 (D) The reaction goes slower as the concentration of I^- is increased.

29. In which ways may a catalyst increase the rate of a reaction?

- I. It may alter the rate law of the reaction.
 II. It may make the overall reaction more exothermic.

- (A) I only (B) II only
 (C) Both I and II (D) Neither I nor II
30. The rate constant of a chemical reaction increases by 63% when the temperature is raised from 40 °C to 45 °C. What is the activation energy of this reaction?
- (A) 1.5 kJ mol⁻¹ (B) 9.7 kJ mol⁻¹
 (C) 76.5 kJ mol⁻¹ (D) 80.9 kJ mol⁻¹

31. Dichromate ion is reduced by chloride ion according to the equation below. If $[\text{Cl}^-]$ is decreasing at a rate of 0.37 mol L⁻¹ min⁻¹, how is the concentration of $\text{Cr}^{3+}(\text{aq})$ changing?

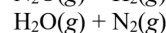
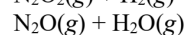


- (A) Increasing by 0.12 mol L⁻¹ min⁻¹
 (B) Increasing by 1.11 mol L⁻¹ min⁻¹
 (C) Decreasing by 0.12 mol L⁻¹ min⁻¹
 (D) Decreasing by 1.11 mol L⁻¹ min⁻¹
32. The isotope cobalt-57 has a half-life of 272 d. How long will it take a sample of ⁵⁷Co to decay to 38% of its original activity?
- (A) 103 d (B) 337 d (C) 188 d (D) 380 d

33. Which statement best describes the variation of the rate constant of a reaction with temperature?

- (A) The rate constant does not change with temperature because it is an unvarying characteristic of the specific reaction.
 (B) The rate constant typically decreases with increasing temperature because fewer molecules are able to adopt the required orientation at higher temperature.
 (C) The rate constant typically increases with increasing temperature because increasing the temperature increases the fraction of collisions that result in reaction.
 (D) The rate constant typically increases with increasing temperature because most reactions become more favorable as the temperature increases.

34. A mechanism for the reaction of nitric oxide with hydrogen to form water and nitrogen gas is proposed below. What rate law is predicted by this mechanism?



- (A) Rate = $k[\text{NO}]^2$
 (B) Rate = $k[\text{NO}]^2[\text{H}_2]$
 (C) Rate = $k[\text{NO}]^2[\text{H}_2]^2$
 (D) Rate = $k \frac{[\text{NO}]^2[\text{H}_2]^2}{[\text{N}_2\text{O}][\text{H}_2\text{O}]}$

35. An irreversible reaction $\text{A} + \text{B} \rightarrow \text{C}$ is carried out under various conditions, but always with $[\text{B}]_0 \gg [\text{A}]_0$. Plots of $1/[\text{A}]$ as a function of time for each reaction are linear, and the slopes of these plots do not depend on the initial concentration of B. What is the rate law for the reaction under these conditions?

- (A) Rate = $k[\text{A}]$ (B) Rate = $k[\text{A}][\text{B}]$
 (C) Rate = $k[\text{A}]^2$ (D) Rate = $k[\text{A}]^2[\text{B}]$

36. The reaction of $\text{H}_2(\text{g})$ with $\text{C}_2\text{H}_4(\text{g})$ to form $\text{C}_2\text{H}_6(\text{g})$ is catalyzed by metallic platinum. Which statement most accurately describes the impact of changing the catalyst in this reaction?

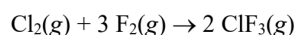
(A) The rate of the reaction would not change if a greater mass of catalyst were used because the platinum is a solid so its concentration is a fixed number.

(B) The rate of the reaction would increase if more finely divided platinum were used because the reaction depends on the surface area of the catalyst.

(C) The ΔH° for the reaction would become more negative if a greater mass of catalyst were used because the catalyst adsorbs the reactants exothermically.

(D) The ΔS° for the reaction would become more negative if more finely divided platinum were used because the catalyst absorbs gas-phase reactants to form ordered surface species.

37. In the reaction



the rate of disappearance of $\text{F}_2(\text{g})$ is 1.0 M s^{-1} . What is the rate of appearance of $\text{ClF}_3(\text{g})$?

(A) 0.33 M s^{-1} (B) 0.67 M s^{-1}
(C) 1.0 M s^{-1} (D) 1.5 M s^{-1}

38. When the rate of the reversible reaction $\text{A} + \text{B} \rightleftharpoons \text{C}$ is studied under a certain set of conditions, it is found that the rate of the forward reaction is $k_f[\text{A}]$. What can be concluded about the rate law for the reverse reaction under these conditions?

(A) $\text{Rate} = k_r[\text{C}]$

(B) $\text{Rate} = k_r \frac{[\text{C}]}{[\text{B}]}$

(C) The rate law of the reverse reaction cannot be determined from the information given.

(D) An error must have been made, since if the reaction is reversible, the forward rate law must be $\text{Rate} = k_f[\text{A}][\text{B}]$.

39. A substance A decomposes irreversibly to form B. A plot of $\ln([\text{A}])$ as a function of time from the beginning of the reaction until A is 97% consumed is a straight line with a negative slope. What is the reaction order in A?

(A) Zero order (B) First order
(C) Second order (D) Third order

40. Which statement about catalysis is correct?

(A) If a catalyst increases the forward rate of a reaction by a factor of two, it must increase the rate of the reverse reaction by a factor of two.

(B) If a catalyst increases the rate of formation of a product by a factor of two, it must increase the rate of formation of the mirror image of the product by a factor of two.

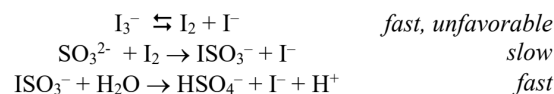
(C) A catalyst must be in the same phase as the reactants and products of the reaction.

(D) A catalyzed reaction must proceed by the same mechanism as the uncatalyzed reaction, but with a lower activation energy.

41. The isotope ^{226}Ra has a half-life for radioactive decay of 1600 y. How long will it take the amount of ^{226}Ra in a sample of $^{226}\text{RaCl}_2$ to decrease by 25%?

(A) 660 y (B) 800 y (C) 1200 y (D) 1600 y

42. The oxidation of sulfite ion by triiodide ion is proposed to take place by the following mechanism:



What rate law is predicted by this mechanism?

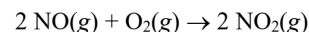
(A) $\text{Rate} = k[\text{I}_3^-]$

(B) $\text{Rate} = k[\text{I}_3^-][\text{SO}_3^{2-}]$

(C) $\text{Rate} = \frac{k[\text{I}_3^-][\text{SO}_3^{2-}]}{[\text{I}^-]}$

(D) $\text{Rate} = \frac{k[\text{I}_3^-][\text{SO}_3^{2-}]}{[\text{H}^+]}$

43. In a study of the reaction below, the concentration of $\text{O}_2(\text{g})$ is found to be decreasing by 0.042 M min^{-1} . At what rate is the concentration of nitrogen dioxide gas changing?



(A) Increasing by 0.021 M min^{-1}

(B) Increasing by 0.042 M min^{-1}

(C) Increasing by 0.084 M min^{-1}

(D) It cannot be determined without knowing the rate law for the reaction.

Kinetics

44. The rate constant for an elementary chemical reaction can be affected by which of the following?

I. Reactant concentrations
II. Product concentrations

- (A) I only (B) II only
(C) Both I and II (D) Neither I nor II

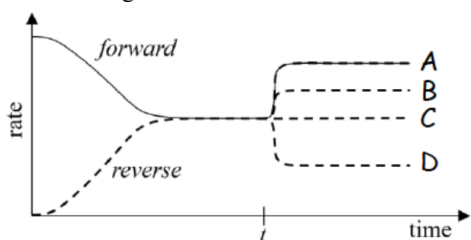
45. Sulfuryl chloride (SO_2Cl_2) decomposes via first-order kinetics. The half-life is 4.1 minutes at a certain temperature. How long does it take for 30% of the SO_2Cl_2 in a sample to decompose at this temperature?

- (A) 0.6 min (B) 2.1 min
(C) 2.5 min (D) 7.1 min

46. The rate constant of a chemical reaction increases 26% when the temperature is raised from 50 °C to 55 °C. What is the activation energy of this reaction?

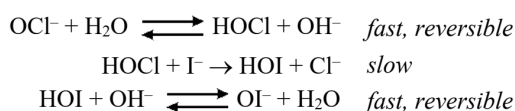
- (A) 1.1 kJ mol⁻¹ (B) 18 kJ mol⁻¹
(C) 41 kJ mol⁻¹ (D) 220 kJ mol⁻¹

47. The graph below shows the forward and reverse rates for a reaction as a function of time. At time t , a catalyst is added to the system, and the forward reaction rate is observed to change as indicated by the solid curve. Which dashed curve best indicates how the reverse reaction rate changes?



- (A) A (B) B (C) C (D) D

48. The following mechanism is proposed for the oxidation of I^- by OCl^- in aqueous solution:



What reaction orders for OCl^- , I^- , and OH^- are consistent with this mechanism?

- | | Order in OCl^- | Order in I^- | Order in OH^- |
|-----|-------------------------|-----------------------|------------------------|
| (A) | 1 | 1 | 1 |
| (B) | 1 | 1 | 0 |
| (C) | 1 | 0 | 0 |
| (D) | 1 | 1 | -1 |

49. The following data were collected for the reaction of A and B:

$[\text{A}]_0$ (M)	$[\text{B}]_0$ (M)	Initial rate (M/s)
0.20	0.20	600
0.60	0.40	7200
0.60	0.80	28800

What is the order of the reaction with respect to A and B?

- (A) 0th order in A, 2nd order in B
(B) 1st order in A, 1st order in B
(C) 2nd order in A, 1st order in B
(D) 1st order in A, 2nd order in B

50. A sample of 0.900 mol N_2O is placed in a sealed container, where it decomposes irreversibly to N_2 and O_2 in a first-order reaction. After 42.0 min, 0.640 mol N_2O remains. How long will it take for the reaction to be 90.0% complete?

- (A) 13.0 min (B) 85.4 min
(C) 131 min (D) 284 min

51. The rate of a reaction is 2.3 times faster at 60 °C than it is at 50 °C. By what factor will the rate increase on going from 60 °C to 70 °C?

- (A) By less than a factor of 2.3
(B) By a factor of 2.3
(C) By more than a factor of 2.3
(D) The rate increase cannot be determined from the information given.

52. How may an enzyme inhibitor decrease the rate of a reaction catalyzed by the enzyme?

I. The inhibitor may bind to the enzyme and block the active site.
II. The inhibitor may shift the equilibrium of the reaction toward the starting materials.

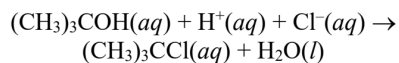
- (A) I only (B) II only
(C) Both I and II (D) Neither I nor II

53. An irreversible reaction, $\text{A} + \text{B} \rightarrow \text{products}$, is studied under conditions where $[\text{B}] \gg [\text{A}]$. When $[\text{B}] = 0.10$ M, a plot of $1/[\text{A}]$ vs. time is linear. When $[\text{B}] = 0.30$ M, the plot of $1/[\text{A}]$ vs. time is indistinguishable from the first plot within experimental error. What is the rate law for this reaction?

- (A) Rate = $k[\text{A}][\text{B}]$ (B) Rate = $k[\text{A}]^2$
(C) Rate = $k[\text{A}]^2[\text{B}]$ (D) Rate = $k[\text{A}][\text{B}]^2$

Kinetics

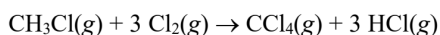
54. *Tert*-butyl alcohol reacts with aqueous hydrochloric acid to give *tert*-butyl chloride.



For this reaction, $\text{Rate} = k[(\text{CH}_3)_3\text{COH}][\text{H}^+]$. With which mechanism is this rate law consistent?

(A)	$(\text{CH}_3)_3\text{COH}(aq) + \text{H}^+(aq) \rightleftharpoons (\text{CH}_3)_3\text{C}(\text{OH}_2)^+(aq)$	<i>fast</i>
	$(\text{CH}_3)_3\text{C}(\text{OH}_2)^+(aq) + \text{Cl}^-(aq) \rightarrow (\text{CH}_3)_3\text{CCl}(aq) + \text{H}_2\text{O}(l)$	<i>slow</i>
(B)	$(\text{CH}_3)_3\text{COH}(aq) + \text{H}^+(aq) \rightleftharpoons (\text{CH}_3)_3\text{C}(\text{OH}_2)^+(aq)$	<i>fast</i>
	$(\text{CH}_3)_3\text{C}(\text{OH}_2)^+(aq) \rightarrow (\text{CH}_3)_3\text{C}^+(aq) + \text{H}_2\text{O}(l)$	<i>slow</i>
	$(\text{CH}_3)_3\text{C}^+(aq) + \text{Cl}^-(aq) \rightarrow (\text{CH}_3)_3\text{CCl}(aq)$	<i>fast</i>
(C)	$(\text{CH}_3)_3\text{COH}(aq) + \text{Cl}^-(aq) \rightarrow (\text{CH}_3)_3\text{CCl}(aq) + \text{OH}^-(aq)$	<i>slow</i>
	$\text{OH}^-(aq) + \text{H}^+(aq) \rightarrow \text{H}_2\text{O}(l)$	<i>fast</i>
(D)	$(\text{CH}_3)_3\text{COH}(aq) \rightarrow (\text{CH}_3)_3\text{C}^+(aq) + \text{OH}^-(aq)$	<i>slow</i>
	$(\text{CH}_3)_3\text{C}^+(aq) + \text{Cl}^-(aq) \rightarrow (\text{CH}_3)_3\text{CCl}(aq)$	<i>fast</i>
	$\text{OH}^-(aq) + \text{H}^+(aq) \rightarrow \text{H}_2\text{O}(l)$	<i>fast</i>

55. Carbon tetrachloride is produced from methyl chloride and chlorine according to the following equation:



If the rate of formation of CCl_4 is measured to be 0.063 M min^{-1} , what is the rate of disappearance of Cl_2 ?

- (A) 0.021 M min^{-1} (B) 0.063 M min^{-1}
 (C) 0.13 M min^{-1} (D) 0.19 M min^{-1}
56. Iodine-131 decays with a half-life of 8.02 d. In a sample initially containing 5.00 mg of ^{131}I , what mass remains after 6.01 d?
- (A) 1.13 mg (B) 1.87 mg
 (C) 2.97 mg (D) 3.13 mg

57. For an irreversible reaction $\text{A} \rightarrow \text{products}$, the graph of $1/[\text{A}]$ as a function of time is linear. What is the reaction order in A?

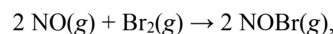
- (A) Zeroth-order
 (B) First-order
 (C) Second-order
 (D) The order in A cannot be determined based on the information given.

58. In comparing two reactions, the reaction with the greater activation energy always has

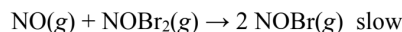
- (A) the slower rate.
 (B) the faster rate.
 (C) the rate that varies less with temperature.
 (D) the rate that varies more with temperature.

Questions 59 and 60 concern the reaction and mechanism below.

The formation of NOBr ,



is studied, and the following mechanism is proposed:



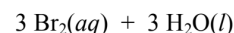
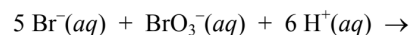
59. What rate law is predicted by this mechanism?

- (A) $\text{Rate} = k[\text{NO}][\text{Br}_2]$ (B) $\text{Rate} = k[\text{NO}]^2[\text{Br}_2]$
 (C) $\text{Rate} = k[\text{NO}][\text{Br}_2]^2$ (D) $\text{Rate} = k[\text{NO}]^2$

60. In this reaction, $\text{NOBr}_2(g)$ is best described as

- (A) an intermediate.
 (B) a product.
 (C) a homogeneous catalyst.
 (D) a heterogeneous catalyst.

61. If elemental bromine is being formed according to the equation below at a rate of 0.056 M s^{-1} , at what rate is bromide ion being consumed?



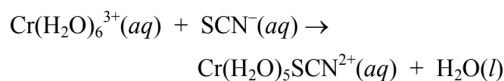
- (A) 0.019 M s^{-1} (B) 0.034 M s^{-1}
 (C) 0.056 M s^{-1} (D) 0.093 M s^{-1}

Kinetics

62. Thallium-201, a radioactive isotope used to image the heart, has a half-life of 3.05 d. How long would it take for a sample of thallium-201 to decay to 18% of its original activity?

(A) 4.4 d (B) 6.1 d (C) 7.5 d (D) 17 d

63. For the reaction:



The following data were collected:

$[\text{Cr}(\text{H}_2\text{O})_6^{3+}]$, M	$[\text{SCN}^-]$, M	Rate, M hr^{-1}
0.028	0.040	8.1×10^{-6}
0.028	0.055	1.1×10^{-5}
0.037	0.055	1.5×10^{-5}

What is the rate law for the reaction?

- (A) $\text{Rate} = (9.1 \times 10^{-9} \text{ M}^{-1} \text{ hr}^{-1})[\text{Cr}(\text{H}_2\text{O})_6^{3+}][\text{SCN}^-]$
 (B) $\text{Rate} = (7.2 \times 10^{-3} \text{ M}^{-1} \text{ hr}^{-1})[\text{Cr}(\text{H}_2\text{O})_6^{3+}][\text{SCN}^-]$
 (C) $\text{Rate} = (2.9 \times 10^{-4} \text{ hr}^{-1})[\text{Cr}(\text{H}_2\text{O})_6^{3+}]$
 (D) $\text{Rate} = (3.9 \times 10^{-4} \text{ hr}^{-1})[\text{Cr}(\text{H}_2\text{O})_6^{3+}]$

64. In which of the following ways may a catalyst increase the rate of a reaction?

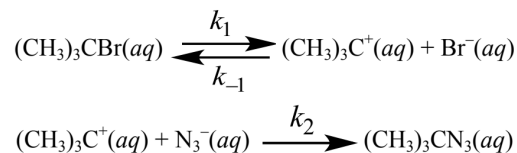
- I. It may alter the rate law.
 II. It may decrease the overall activation energy.

(A) I only (B) II only
 (C) Both I and II (D) Neither I nor II

65. For an exothermic reaction, which of the following best describes the effect of increasing the temperature on the forward and reverse reactions?

- (A) Both the forward and reverse rates increase, but the forward rate increases more than the reverse rate.
 (B) Both the forward and reverse rates increase, but the reverse rate increases more than the forward rate.
 (C) The forward rate increases while the reverse rate decreases.
 (D) The reverse rate increases while the forward rate decreases.

66. The reaction of *tert*-butyl bromide with azide ion in aqueous solution is proposed to proceed by the following mechanism:



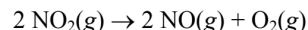
Assuming that $(\text{CH}_3)_3\text{C}^+(\text{aq})$ achieves a steady-state concentration, but making no further assumptions about the relative magnitudes of the three rate constants, what is the rate law for this reaction?

- (A) $\text{Rate} = k_1[(\text{CH}_3)_3\text{CBr}]$
 (B) $\text{Rate} = k_2[(\text{CH}_3)_3\text{CBr}][\text{N}_3^-]$
 (C) $\text{Rate} = \frac{k_1 k_2 [(\text{CH}_3)_3\text{CBr}][\text{N}_3^-]}{k_{-1}[\text{Br}^-]}$
 (D) $\text{Rate} = \frac{k_1 k_2 [(\text{CH}_3)_3\text{CBr}][\text{N}_3^-]}{k_{-1}[\text{Br}^-] + k_2[\text{N}_3^-]}$

67. Which statement about chemical reaction mechanisms is correct?

- (A) The overall rate law can be determined from any step in the mechanism.
 (B) The rate of a reaction is the rate of the fastest elementary step of its mechanism.
 (C) The chemical equation for the sum of all the elementary steps is the chemical equation of the overall reaction.
 (D) Species that are produced and subsequently consumed in the mechanism serve as catalysts for the reaction.

68. At 280 °C, nitrogen dioxide decomposes to nitric oxide and oxygen:

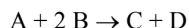


In one experiment, the concentration of NO_2 decreased from 0.0100 M to 0.0050 M over the course of 100. s. What was the average rate of disappearance of $\text{NO}_2(\text{g})$?

- (A) $1.0 \times 10^{-2} \text{ M s}^{-1}$ (B) $1.0 \times 10^{-4} \text{ M s}^{-1}$
 (C) $5.0 \times 10^{-5} \text{ M s}^{-1}$ (D) $2.5 \times 10^{-5} \text{ M s}^{-1}$

Kinetics

69. What is the rate law for the following reaction?



$[\text{A}]_0, \text{M}$	$[\text{B}]_0, \text{M}$	Initial rate, M s^{-1}
0.050	0.100	0.085
0.050	0.200	0.170
0.100	0.300	0.510

- (A) $\text{Rate} = k[\text{A}]$ (B) $\text{Rate} = k[\text{B}]$
(C) $\text{Rate} = k[\text{A}][\text{B}]$ (D) $\text{Rate} = k[\text{A}][\text{B}]^2$
70. What are the units of k if the rate law of a reaction is $\text{rate} = k[\text{X}]^0[\text{Y}]^0$?
- (A) M s^{-1} (B) s^{-1}
(C) $\text{M}^{-1} \text{s}^{-1}$ (D) k is dimensionless
71. Hydrogen peroxide, $\text{H}_2\text{O}_2(aq)$, decomposes into water and oxygen. Adding a small amount of $\text{FeCl}_3(aq)$ increases the rate of gas evolution in this reaction. What is the best description of the role of FeCl_3 ?
- (A) Transition state
(B) Reaction intermediate
(C) Heterogeneous catalyst
(D) Homogeneous catalyst
72. A substance decomposes in a first-order reaction with a rate constant of $6.70 \times 10^{-4} \text{s}^{-1}$. If the initial concentration of the substance is 1.50 M, what is its concentration after 500. s?
- (A) 1.07 M (B) 0.503 M
(C) 0.335 M (D) 0.128 M